

- 8 (a) Sampling is used to find out the variety of species in an ecosystem and the size of the population of each species.

A study was carried out to investigate the biodiversity of two fields, **A** and **B**.

Field **A** had **not** been used for growing crops for 10 years. Field **B** had been used for growing crops until one year before the study. Random sampling was carried out.

Describe how random sampling could be carried out on plant species in the two fields.

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..... [4]

- (b) Table 8.1 shows the results of the sampling of field **A**.

Table 8.1

species	number of individuals (n)
<i>Scabiosa columbaria</i>	13
<i>Centaurea centaurium</i>	15
<i>Primula veris</i>	26
<i>Trifolium pratense</i>	36
<i>Leucanthemum vulgare</i>	11
<i>Silybum marianum</i>	5
<i>Anacamptis morio</i>	8
total number of individuals (N)	114

Simpson's Index of Diversity (D) is a value of the species diversity in an area.

The formula for this is:

$$D = 1 - \left(\sum \left(\frac{n}{N} \right)^2 \right)$$

Σ = sum of

n = number of individuals of a species

N = number of individuals of all species

(i) Complete Table 8.2 for *Primula veris*.

Table 8.2

species	n	$\frac{n}{N}$	$\left[\frac{n}{N} \right]^2$
<i>Scabiosa columbaria</i>	13	0.11	0.01
<i>Centaurea centaurium</i>	15	0.13	0.02
<i>Primula veris</i>	26		
<i>Trifolium pratense</i>	36	0.32	0.10
<i>Leucanthemum vulgare</i>	11	0.10	0.01
<i>Silybum marianum</i>	5	0.04	0.00
<i>Anacamptis morio</i>	8	0.07	0.00

[1]

(ii) Use the data in Table 8.2 to calculate Simpson's Index of Diversity (D) for field **A**.

Show your working and write your answer to **two** decimal places.

$D =$ [2]

(iii) The value of D for field **B** was 0.45.

Comment on the values of D for both fields.

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 [1]

[Total: 8]

Section B

Answer **one** question.